Outdoor appliance cabinet: fridge + stacked washer/dryer





A square black formply box in the IKEA GLAMBERGET style: flush top, flush head rail, two inset top-hung sliding doors, cheeks notched to read as slab legs, and a full-height American walnut stile on each door as the handle.

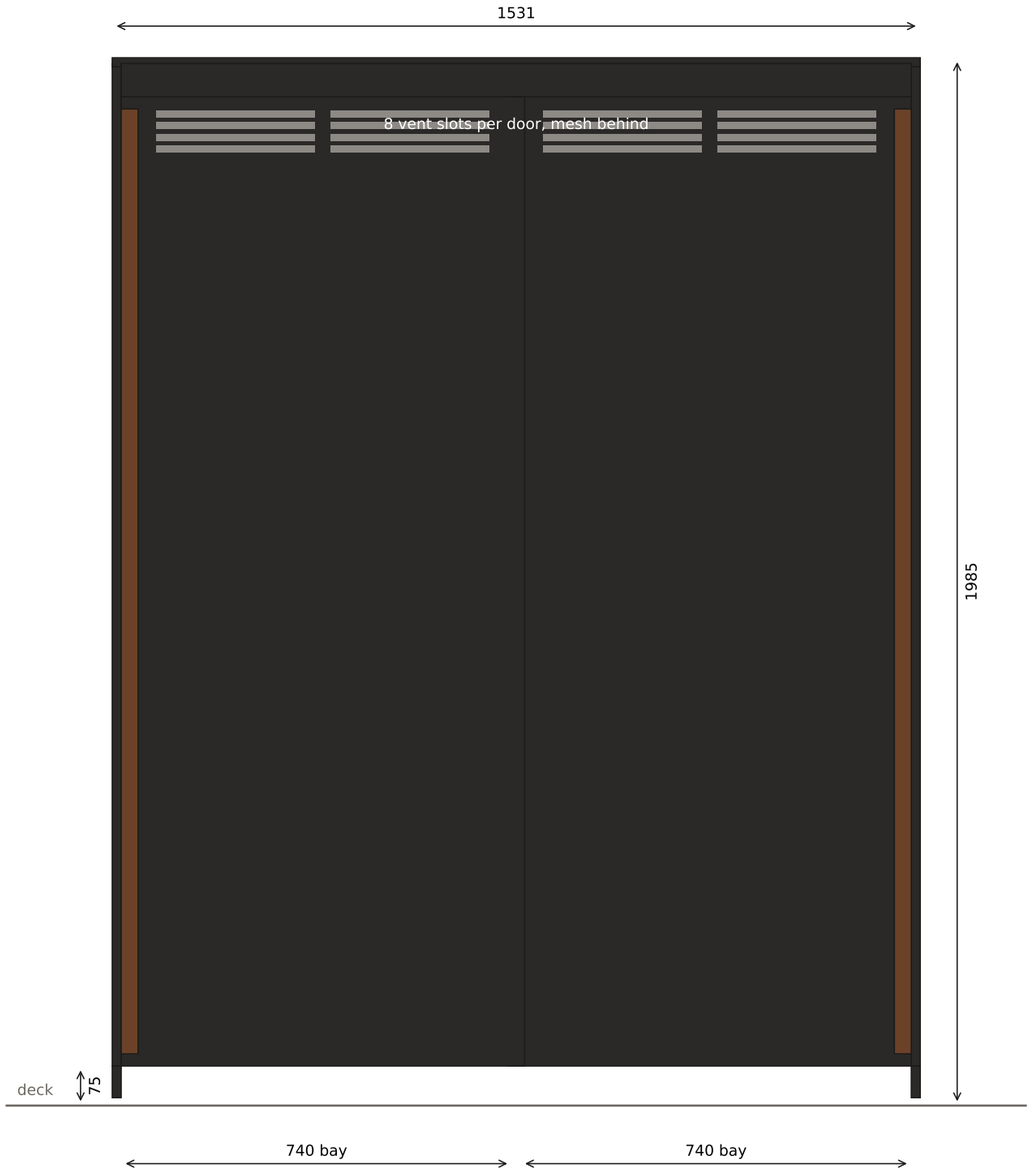
1531 W × 900 D × 1985 H. Two 740 clear bays. 9 CNC parts on 8 sheets of 2400 × 1200 × 17.

Cool air enters through the 75 gap under the doors; warm air leaves through the meshed slots high in each door. Solid backs. Every panel stands on ASA feet 15 above the deck. The appliances stand on the deck; the cabinet carries none of them.

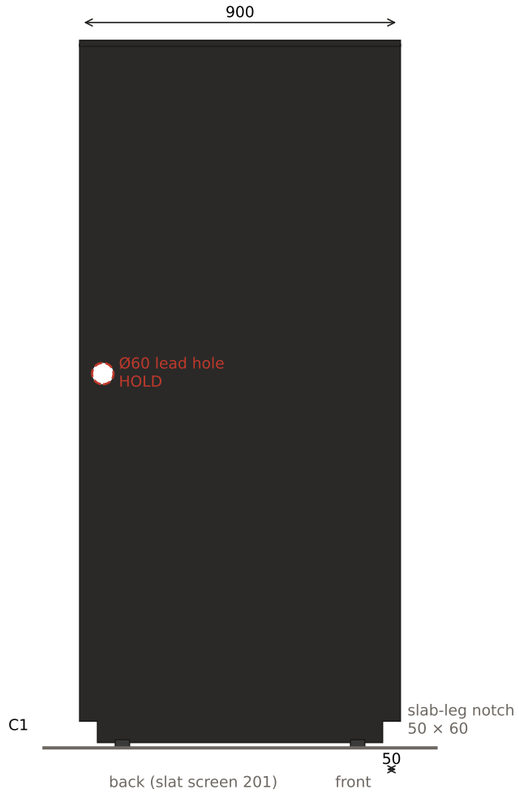
17 of 17 design checks pass (sheet 6). Open items must be closed before the cut.

General arrangement

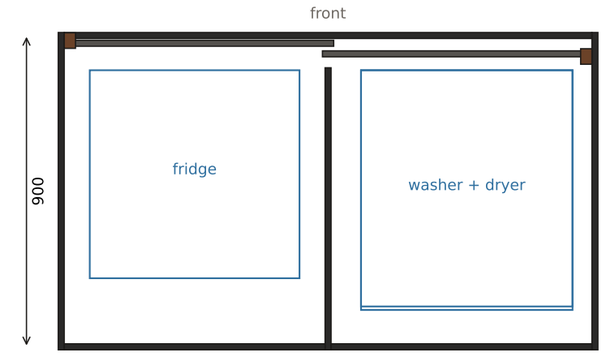
Front elevation (doors closed)



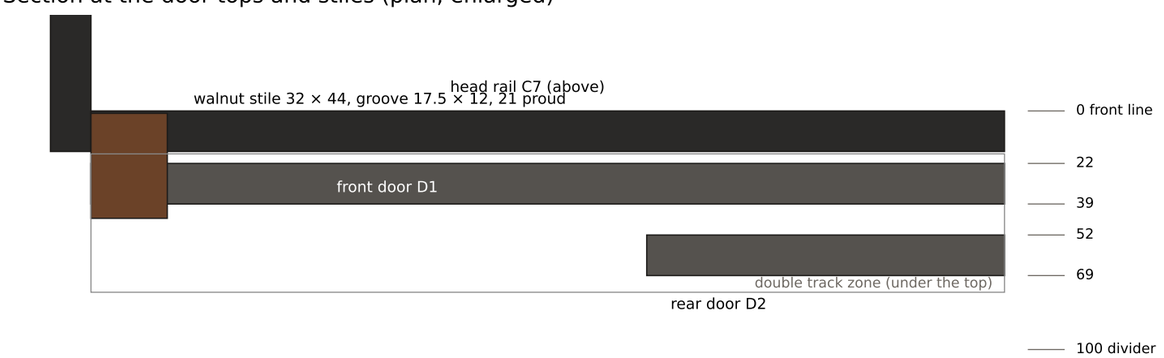
Side elevation (studio cheek, outside)



Plan (top removed)

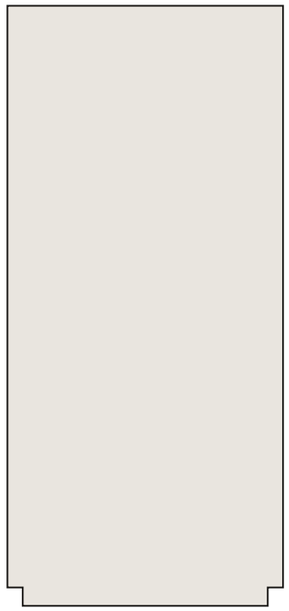


Section at the door tops and stiles (plan, enlarged)



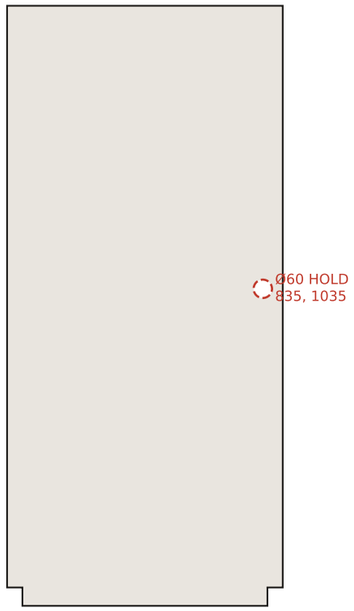
CNC part drawings

C1 Left cheek 900 × 1959 ×1



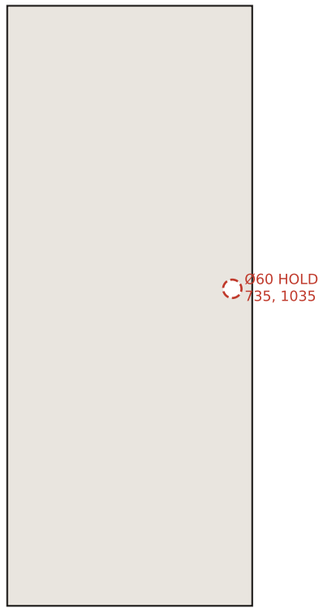
X = 0 at the cabinet front edge, Y = 0 at the bottom edge. Slab-leg notches 50 x 60 at both bottom corners.

C3 Studio cheek 900 × 1959 ×1



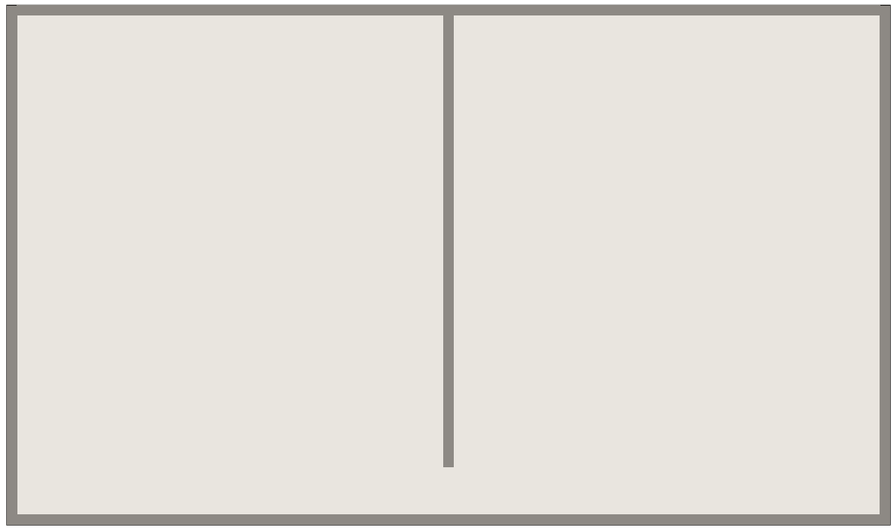
X = 0 at the cabinet front edge (mirror view: routed from the outside face). Slab-leg notches as C1. HOLD hole: confirm on site.

C2 Divider 800 × 1959 ×1



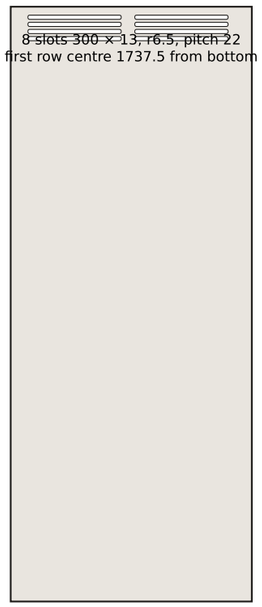
X = 0 at the front edge. HOLD hole: confirm on site.

C4 Cap 1531 × 900 ×1



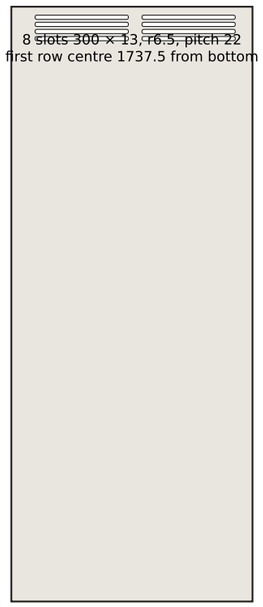
Routed on the UNDERSIDE: 6 mm housings for every vertical and the head rail. X = 0 at the kitchen-side edge, Y = 0 at the front edge.

D1 Fridge door 744.5 × 1836 ×1



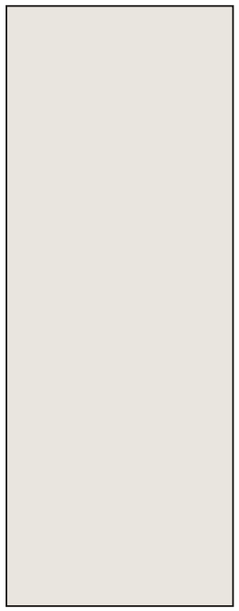
Slots: 12 mm cutter, round ends. The walnut stile takes the left edge.

D2 Laundry door 744.5 × 1836 ×1



Slots: 12 mm cutter, round ends. The walnut stile takes the right edge.

C5 Back, fridge bay 740 × 1959 ×1



C7 Head rail 1497 × 76 ×1

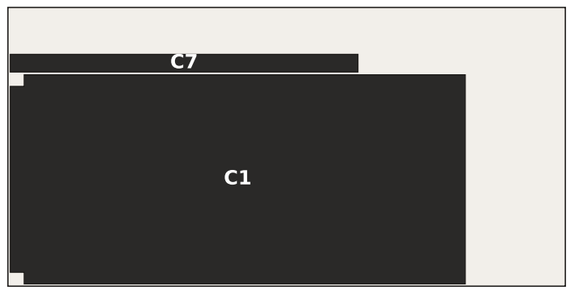


C6 Back, laundry bay: as C5.
C4 is routed on its UNDERSIDE: 6 deep housings (grey) for C1, C2, C3, the backs and the head rail.
Cheek housings are 0.6 over the panel for glue and fit.

Cut list and indicative nests

Code	Part	Size	Qty	Notes
C1	Left cheek	900 × 1959	1	X = 0 at the cabinet front edge, Y = 0 at the bottom edge. Slab-leg notches 50 x 60 at both bottom corners.
C2	Divider	800 × 1959	1	X = 0 at the front edge. HOLD hole: confirm on site.
C3	Studio cheek	900 × 1959	1	X = 0 at the cabinet front edge (mirror view: routed from the outside face). Slab-leg notches as C1. […]
C4	Cap	1531 × 900	1	Routed on the UNDERSIDE: 6 mm housings for every vertical and the head rail. X = 0 at the kitchen-side […]
C5	Back, fridge bay	740 × 1959	1	
C6	Back, laundry bay	740 × 1959	1	
C7	Head rail	1497 × 76	1	
D1	Fridge door	744.5 × 1836	1	Slots: 12 mm cutter, round ends. The walnut stile takes the left edge.
D2	Laundry door	744.5 × 1836	1	Slots: 12 mm cutter, round ends. The walnut stile takes the right edge.

S1 2400 × 1200 × 17



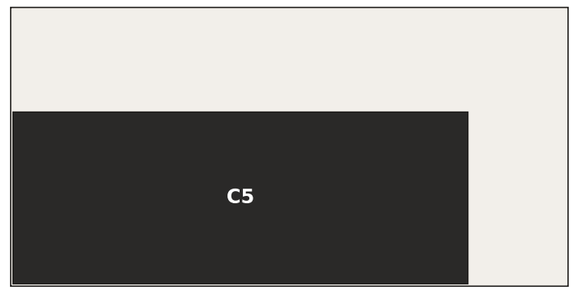
S2 2400 × 1200 × 17



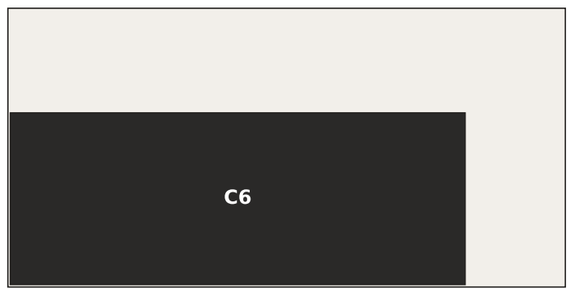
S3 2400 × 1200 × 17



S4 2400 × 1200 × 17



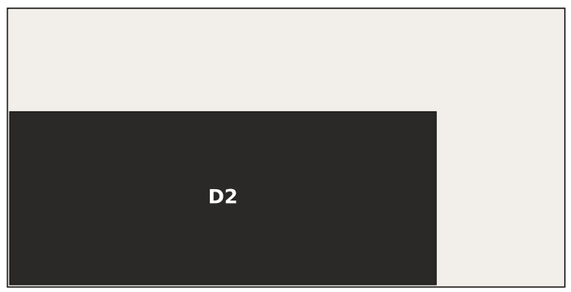
S5 2400 × 1200 × 17



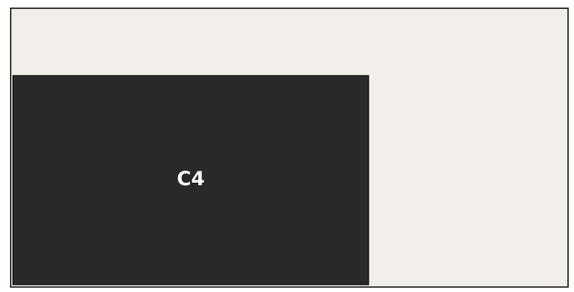
S6 2400 × 1200 × 17



S7 2400 × 1200 × 17



S8 2400 × 1200 × 17



Build steps and shopping list

1. Confirm before cutting

Confirm the washer and dryer models and their stacking kit (dryer must be heat-pump or condenser). Electrician confirms three accessible outlets on the studio wall. Measure the deck, the 201 gap to the slat screen and the 195 gap to the studio wall. Release the two HOLD lead holes, or leave them off the cut.

2. Seal and print

Seal every CNC-cut edge with two coats of exterior edge sealer, paying most attention to bottom edges and the vent slots. Print the ASA parts (Prints tab).

3. Feet on the deck

Mark the footprint on the deck. Screw the ten ASA feet down with 10g x 40 stainless: two under each cheek, two under the divider, four under the backs. Check they are in one plane with a straightedge; pack under a foot with a shim if a board is low.

4. Carcass on the cap, upside down

Lay the top film-face down on padded trestles, housings up. Run PU sealant in each housing. Stand C1, C2, C3, C5 and C6 in their housings; the housings square the box. Screw down through the top into each edge, 8g x 50 stainless at 200 centres, counter-sunk and sealed.

5. Back corner angles

Fit the four 25 x 25 x 3 black aluminium angles inside the back corners, sealant in the joint, 8g x 25 stainless at 200 both legs.

6. Stand it up

Two people. Turn the carcass upright and lower the panel edges into the feet. Check plumb and diagonals, then screw through each foot into the panel (6g x 20 stainless).

7. Head rail and track

Push the head rail up into its housing between the cheeks, flush with their fronts, and screw down through the top. Cut the double top track to 1497 and screw it up into the top behind the head rail.

8. Walnut stiles and doors

Machine the two walnut stiles (32 wide, groove 17.5 x 12, finger-jointed or scarfed from offcuts if needed). Oil all faces and seal the end grain. PU-glue each stile onto its door edge and screw through the groove cheeks. Fit the insect mesh behind the high vent groups. Hang D1 on the front track and D2 on the rear; set both doors 75 off the deck.

9. Floor guide

Close both doors. Screw the ASA floor guide to the deck where they cross so the fins straddle each door; run both doors end to end and check the stiles pass over the fins.

10. Grommets and leads

Only after the HOLD is released: fit the split grommets in the lead holes and route each original lead separately to the wall. No powerboard, nothing in a door track or under a machine foot.

11. Appliances in

Roll the fridge in, door hinge on the left, 107 back from the cabinet front. Stand the washer, fit the maker's stacking kit and lift the dryer on. Level all three on their own feet; the cabinet carries none of them.

Type	Item	Qty	Use
Sheet	Black film-faced formply 2400 x 1200 x 17, exterior (A-bond)	8 sheets	CNC supplier supplies and cuts (Cut list tab)
Sheet	Exterior edge sealer, black	1 L	Two coats on every cut edge before assembly
Sheet	PU sealant/adhesive, black (Sikaflex-11FC or similar)	4 x 300 ml	Housings, angles, mesh, stiles
Hardware	Double top-hung exterior sliding track kit, 2 leaves to 40 kg each	1 kit, cut to 1497	Stainless or anodised wheels and hangers
Hardware	Aluminium angle 25 x 25 x 3, black anodised	4 x 1850	Inside the back corners
Hardware	Stainless insect mesh	2 m x 300	Behind the two high vent groups
Timber	American walnut, 32 x 44 finished, from workshop offcuts	2 x 1800	Door edge stiles (handles); finger-joint or scarf if short
Timber	Exterior hardwood oil (e.g. Cutek or Intergrain UltraDeck Oil)	500 ml	Walnut stiles; end-grain sealer for the ends
Fixings	8g x 50 stainless countersunk timber screws	60	Through the cap into the panel edges
Fixings	8g x 25 stainless countersunk	80	Corner angles
Fixings	10g x 40 stainless countersunk	24	Feet and floor guide into the deck
Fixings	6g x 20 stainless pan	30	Feet into panels
Fixings	8g x 30 stainless countersunk	24	Through the stile cheeks into the door edges
Fixings	Stainless staples, 10 mm	1 box	Mesh
Print	ASA filament, black	about 0.8 kg	Prints tab: 3 STLs, 13 prints

Checks, open items and ASA prints

	Check	Result
Pass	Fridge side clearance (maker 50)	72.5 each side
Pass	Fridge back clearance (maker 50)	186 to the back panel
Pass	Fridge top clearance (maker 100)	183 to the cap; track and head rail are in front of the fridge face
Pass	Stack under the cap	238 above the dryer top
Pass	Fridge passes the open leaf	opening 17-749.5, fridge 89.5-684.5
Pass	Laundry passes the open leaf	opening 841.5-1514, machines 859.0-1459.0
Pass	Rear stile clear of the closed front door	28 mm finger gap when the laundry door is open
Pass	Stiles pass the other door	7 and 7 mm running gaps
Pass	Front stile inside the cheek faces	1 behind the front line
Pass	Rear stile clear of the divider	10 mm
Pass	Stiles clear the floor guide	stiles start 8 above the guide fins
Pass	Door tops hidden behind the head rail	13 mm overlap
Pass	Low inlet under the doors	1123 cm2 open gap under the doors
Pass	High vents above the appliances	lowest high slot 1806, fridge 1785, stack 1730
Pass	Doors clear of the deck	75 above the deck
Pass	Floor guide engages both doors	fins lap 15 mm up each door
Pass	Fridge door opens 115 deg past the cheek	closest 24 mm at 115 deg (simplified slab, left hinge)

Open item	Before the cut
Washer and dryer models	Provisional 600 x 680 x 850 and 600 x 670 x 850. Confirm both models and the maker's stacking kit before cutting; the dryer must be heat-pump or condenser.
Power	Owner says there is a power point on the studio wall. Position, outlet count and weather cover are unmeasured. The two lead holes are HOLD until an electrician confirms.
Weather	The pergola is open, so the cabinet takes rain. Film faces are weatherproof; cut edges are the weak point. Seal every edge twice and re-seal any chip. The top is flush (GLAMBERGET look), so its four edges carry the weather: seal them twice and keep an eye on them. Every panel stands 15 off the deck.
Track kit	Choose an exterior double top-hung kit rated for two 764 x 1836 doors (about 20 kg each with the stile). It must fit the 45 high x 58 deep zone behind the 70 mm head rail.
Washer door	With the laundry door open, the parked door and its walnut stile sit 17 mm left of the washer. A left-hinged washer door opened past 90° will touch the stile. Prefer a right-hinged washer, or open it to 90° only.
Walnut outdoors	American walnut is only moderately durable outdoors. Under the open pergola it will grey and needs re-oiling once or twice a year. Seal the end grain at the bottom of each stile.
Site measure	Footprint, the 201 gap to the slat screen and the 195 gap to the studio wall come from the handover setout. Measure on site before the cut is released.
Fridge door	Shown left-hinged. If it is re-hung to the right, re-check the swing: it would open towards the divider.

ASA print	Qty	Size	Notes
03_bypass_floor_guide.stl	1	70 × 74 × 90	Floor guide at the centre of the opening, where the leaves cross. Three fins rise 90 above the deck and lap 15 mm up each door (door bottoms are 75 clear of the deck), holding both doors on their lines. The walnut stiles start just above the fins so they pass over. 4 x 10g x 40 stainless into the deck. It sits clear of both appliance paths.
04_cable_grommet.stl	2	82 × 82 × 20	Split grommet for the two 60 mm lead holes (C2, C3). Smooth bore protects the leads. Print flange down. HOLD: confirm the plugs fit through 48 mm or enlarge the hole first.
05_panel_foot.stl	10	80 × 40 × 23	Foot under the ply: the panel edge sits in its 8 mm seat with its underside 15 mm above the deck so the ply never sits in water and the deck drains. 2 x 10g x 40 stainless into the deck. Print 100% infill on the bottom 3 mm, 40% gyroid above.